Testovanie odlišnosti generovaných riešení úloh s ováhovanými akciami

Testovanie vykonávam pomocou programov *hopper_cost_evaluator.py* a *plan_edit_distance.py*. Všetky úlohy používajú doménu *HopperCostDomain.pddl*.

Pomocou *hopper_cost_evaluator.py* spúšťam plánovač 100 krát na tej istej doméne a probléme. Po každej iterácii sa ceny akcií zmenia podľa použitej funkcie. Postupne budem testovať tri funkcie:

1. Za prvý výskyt akcie v nájdenom riešení úlohy, zvýšim jej cenu o jedna
2. Za každý výskyt akcie v nájdenom riešení úlohy, zvýšim jej cenu o jedna
3. Za prvý výskyt akcie v nájdenom riešení úlohy, zvýšim jej cenu o náhodnú hodnotu medzi 0 (vrátane) a 5 (5 už nie)

Po získaní všetkých riešení úlohy pomocou vybranej funkcie, porovnám editačnú vzdialenosť týchto riešení pomocou *plan_edit_distance.py*.

1. Úloha – Prines starú knihu (*HopperCostBringOldBook.pddl*)

1. Funkcia

```
e = HopperCostEvaluator()
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostBringOldBook.pddl", 100)
```

Nájdené riešenia:

***1. solution**:*
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
*Solution was repeated **72 times**.*

***2. solution**:*
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(attack_character hopper mage mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)
*Solution was repeated **1 times**.*

3. solution:
(steal hopper mage gold_coin mill)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
*Solution was repeated **2 times**.*

***4. solution**:*
(go hopper mill village)
(go hopper village meadow)
(collect hopper black_lily meadow)
(go hopper meadow village)
(go hopper village shop)
(buy shopkeeper hopper old_book black_lily shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
*Solution was repeated **1 times**.*

***5. solution**:*
(attack_character hopper mage mill)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)
*Solution was repeated **24 times**.*

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	2., 3.	2., 4.	2., 5.	3., 4.	3., 5.	4., 5.	**Priemer**
Editačná vzdialenosť	7	2	4	7	7	8	2	5,34	6	8,34	**5,67**

2. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(2)
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostBringOldBook.pddl", 100)
```

Nájdené riešenia:

1. solution:
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **80 times**.

2. solution:
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(attack_character hopper mage mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)
Solution was repeated **1 times**.

3. solution:
(steal hopper mage gold_coin mill)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
Solution was repeated **3 times**.

4. solution:
(go hopper mill village)
(go hopper village meadow)
(collect hopper black_lily meadow)
(go hopper meadow village)
(go hopper village shop)
(buy shopkeeper hopper old_book black_lily shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **2 times**.

5. solution:
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **1 times**.

6. solution:
(attack_character hopper mage mill)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)
Solution was repeated **13 times**.

Nové je riešenie číslo 5. Ostatné riešenia boli nájdené aj predchádzajúcou funkciou.

Editačná vzdialenosť riešení:

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	1., 6.	2., 3.	2., 4.	2., 5.	2., 6.	3., 4.
Editačná vzdialenosť	7	2	4	3	7	7	8	4	2	5,34
Porovnávané riešenia	3., 5.	3., 6.	4., 5.	4., 6.	5. , 6.	**Priemer**				
Editačná vzdialenosť	5	6	4,34	8,34	4	**5,13**				

3. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(3)
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostBringOldBook.pddl", 100)
```

1. solution:
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **55 times**.

2. solution:
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **15 times**.

3. solution:
(steal hopper mage gold_coin mill)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
Solution was repeated **19 times**.

4. solution:
(go hopper mill village)
(go hopper village meadow)
(collect hopper black_lily meadow)
(go hopper meadow village)
(go hopper village shop)
(buy shopkeeper hopper old_book black_lily shop)
(go hopper shop village)
(go hopper village mill)
(buy mage hopper gold_coin old_book mill)
Solution was repeated **4 times**.

5. solution:
(attack_character hopper mage mill)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
(kill hopper mage mill)
(take hopper mage gold_coin mill)
Solution was repeated **4 times**.

6. solution:
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(go hopper shop village)
(go hopper village mill)
(steal hopper mage gold_coin mill)
(go hopper mill village)
(go hopper village shop)
(take hopper shopkeeper old_book shop)
(go hopper shop village)
(go hopper village mill)
(player_give hopper mage old_book mill)
Solution was repeated **3 times**.

Riešenie 6 je nové, ešte sa neobjavilo v predchádzajúcich prípadoch. Chýba riešenie 2 z predchádzajúcej funkcie.

Editačná vzdialenosť riešení:

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	1., 6.	2., 3.	2., 4.	2., 5.	2., 6.	3., 4.
Editačná vzdialenosť	3	2	4	7	8,75	5	4,33	4	6	5,34
Porovnávané riešenia	3., 5.	3., 6.	4., 5.	4., 6.	5. , 6.	**Priemer**				
Editačná vzdialenosť	6	7	8,33	7,67	8	**5,76**				

2. Úloha – Získaj informáciu (*HopperCostLearnAboutNote.pddl*)

1. Funkcia

```
e = HopperCostEvaluator()
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostLearnAboutNote.pddl", 100)
```

Nájdené riešenia:

1. solution:
(go hopper mill village)
(spy hopper hunter note village)
Solution was repeated **40 times**.

2. solution:
(go hopper mill village)
(talk hunter hopper note village)
Solution was repeated **40 times**.

3. solution:
(go hopper mill village)
(go hopper village shop)
(collect hopper list shop)
(read hopper list note)
Solution was repeated **17 times**.

4. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(give shopkeeper hopper list shop)
(read hopper list note)
Solution was repeated **1 times**.

5. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **1 times**.

6. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **1 times**.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	1., 6.	2., 3.	2., 4.	2., 5.	2., 6.	3., 4.
Editačná vzdialenosť	1	3	4	4	6	3	4	4	6	2

Porovnávané riešenia	3., 5.	3., 6.	4., 5.	4., 6.	5. , 6.	**Priemer**
Editačná vzdialenosť	2	4	1	3	3	**3,33**

2. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(2)
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostLearnAboutNote.pddl", 100)
```

Nájdené riešenia:

1. solution:
(go hopper mill village)
(spy hopper hunter note village)
Solution was repeated **40 times**.

2. solution:
(go hopper mill village)
(talk hunter hopper note village)
Solution was repeated **40 times**.

3. solution:
(go hopper mill village)
(go hopper village shop)
(collect hopper list shop)
(read hopper list note)
Solution was repeated **17 times**.

4. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(give shopkeeper hopper list shop)
(read hopper list note)
Solution was repeated **1 times**.

5. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **1 times.**

6. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **1 times**.

Presne tie isté riešenia, ako pri použití prvej funkcie.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	1., 6.	2., 3.	2., 4.	2., 5.	2., 6.	3., 4.
Editačná vzdialenosť	1	3	4	4	6	3	4	4	6	2
Porovnávané riešenia	3., 5.	3., 6.	4., 5.	4., 6.	5. , 6.	**Priemer**				
Editačná vzdialenosť	2	4	1	3	3	**3,33**				

3. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(2)
```

```
e.run("D:\...\hopperCostDomain.pddl","D:\...\hopperCostLearnAboutNote.pddl", 100)
```

Nájdené riešenia:

1. solution:
(go hopper mill village)
(spy hopper hunter note village)
Solution was repeated **38 times**.

2. solution:
(go hopper mill village)
(talk hunter hopper note village)
Solution was repeated **39 times**.

3. solution:
(go hopper mill village)
(go hopper village shop)
(collect hopper list shop)
(read hopper list note)
Solution was repeated **19 times**.

4. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(give shopkeeper hopper list shop)
(read hopper list note)
Solution was repeated **1 times**.

5. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(steal hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **2 times**.

6. solution:
(collect shopkeeper list shop)
(go hopper mill village)
(go hopper village shop)
(attack_character hopper shopkeeper shop)
(kill hopper shopkeeper shop)
(take hopper shopkeeper list shop)
(read hopper list note)
Solution was repeated **1 times**.

Presne tie isté riešenia, ako pri použití prvej a druhej funkcie.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	1., 5.	1., 6.	2., 3.	2., 4.	2., 5.	2., 6.	3., 4.
Editačná vzdialenosť	1	3	4	4	6	3	4	4	6	2
Porovnávané riešenia	3., 5.	3., 6.	4., 5.	4., 6.	5. , 6.	**Priemer**				
Editačná vzdialenosť	2	4	1	3	3	**3,33**				

3. Úloha – Problém s vlkolakom (*hopperCostWerewolfProblem.pddl*)

1. Funkcia

```
e = HopperCostEvaluator()
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostWerewolfProblem.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(kill hopper werewolf meadow)
Solution was repeated **50 times**.

2. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(scare_of hopper werewolf meadow)
(run_away werewolf meadow forest)
Solution was repeated **50 times**.

Editačná vzdialenosť riešení:

Porovnávané riešenia	1., 2.	**Priemer**
Editačná vzdialenosť	2	**2**

2. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(2)
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostWerewolfProblem.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(kill hopper werewolf meadow)
Solution was repeated **50 times**.

2. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(scare_of hopper werewolf meadow)
(run_away werewolf meadow forest)
Solution was repeated **50 times**.

2. funkcia nenašla žiadne nové riešenie.

Editačná vzdialenosť riešení:

Porovnávané riešenia	1., 2.	**Priemer**
Editačná vzdialenosť	2	**2**

2. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(3)
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostWerewolfProblem.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(kill hopper werewolf meadow)
Solution was repeated **48 times**.

2. solution:
(go hopper shop village)
(go hopper village meadow)
(attack_character hopper werewolf meadow)
(scare_of hopper werewolf meadow)
(run_away werewolf meadow forest)
Solution was repeated **52 times**.

Zdá sa, že pre túto úlohu existujú len tieto dve riešenia.

Editačná vzdialenosť riešení:

Porovnávané riešenia	1., 2.	**Priemer**
Editačná vzdialenosť	2	**2**

4. Úloha – Vylieč lovca (*hopperCostCureHunter.pddl*)

1. Funkcia

```
e = HopperCostEvaluator()
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostCureHunter.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper village mill)
(steal hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **35 times**.

2. solution:
(give hunter hopper gold village)
(go hopper village mill)
(buy mage hopper med2 gold mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **18 times**.

3. solution:
(go hopper village meadow)
(go hopper meadow forest)
(collect hopper med1 forest)
(go hopper forest meadow)
(go hopper meadow village)
(cure hopper hunter med1 village)
Solution was repeated **35 times**.

4. solution:
(go hopper village mill)
(attack_character hopper mage mill)
(kill hopper mage mill)
(take hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **12 times**.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	2., 3.	2., 4.	3., 4.	**Priemer**
Editačná vzdialenosť	2	3,92	3	4,25	4	3,912	**3,51**

2. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(2)
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostCureHunter.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper village mill)
(steal hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **35 times**.

2. solution:

(give hunter hopper gold village)
(go hopper village mill)
(buy mage hopper med2 gold mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **18 times**.

3. solution:
(go hopper village meadow)
(go hopper meadow forest)
(collect hopper med1 forest)
(go hopper forest meadow)
(go hopper meadow village)
(cure hopper hunter med1 village)
Solution was repeated **35 times**.

4. solution:
(go hopper village mill)
(attack_character hopper mage mill)
(kill hopper mage mill)
(take hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **12 times.**

Rovnaké riešenia ako v prvom prípade.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	2., 3.	2., 4.	3., 4.	**Priemer**
Editačná vzdialenosť	2	3,92	3	4,25	4	3,912	**3,51**

3. Funkcia

```
e = HopperCostEvaluator()
e.setActionCostFunction(3)
e.run("D:\...\hopperCostDomain.pddl","D:\...\ hopperCostCureHunter.pddl ", 100)
```

Nájdené riešenia:

1. solution:
(go hopper village mill)
(steal hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **30 times**.

2. solution:
(give hunter hopper gold village)
(go hopper village mill)
(buy mage hopper med2 gold mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **19 times**.

3. solution:
(go hopper village meadow)
(go hopper meadow forest)
(collect hopper med1 forest)
(go hopper forest meadow)
(go hopper meadow village)
(cure hopper hunter med1 village)
Solution was repeated **37 times**.

4. solution:
(go hopper village mill)
(attack_character hopper mage mill)
(kill hopper mage mill)
(take hopper mage med2 mill)
(go hopper mill village)
(cure hopper hunter med2 village)
Solution was repeated **14 times**.

Rovnaké riešenia ako v prvom prípade.

Editačná vzdialenosť riešení:

```
readAndCompare("foundSolutions.txt")
```

Porovnávané riešenia	1., 2.	1., 3.	1., 4.	2., 3.	2., 4.	3., 4.	**Priemer**
Editačná vzdialenosť	2	3,92	3	4,25	4	3,912	**3,51**

Záver

Ukázalo sa, že vo väčšine prípadov nezáleží na tom, ktorú z troch funkcií zvyšovania ceny akcií použijem. V prvej úlohe našli 2. a 3. funkcia o jedno riešenie viac, ako 1. funkcia. V ostatných prípadoch však boli získané riešenia rovnaké.

Skúsim ešte vykonať jeden test. Pre každú úlohu skúsim hľadať riešenia postupne so všetkými funkciami, ale pri menšom počte iterácií. Teraz som mala 100. Vyskúšam spraviť 10 iterácií a skúsim sa pozrieť, či je nejaký rozdiel v tom koľko iterácií funkcie potrebujú, aby našli všetky riešenia, ktoré našli pri 100 iteráciách.

10 iterácií:

Úloha	**1. Funkcia**	**2. Funkcia**	**3. Funkcia**	**Počet riešení pri 100 iteráciách**
Prines starú knihu	5	5	6	6
Získaj informáciu	4	4	3	6
Problém s vlkolakom	2	2	2	2
Vylieč lovca	4	4	4	4

20 iterácií:

Úloha	**1. Funkcia**	**2. Funkcia**	**3. Funkcia**	**Počet riešení pri 100 iteráciách**
Prines starú knihu	5	6	5	6
Získaj informáciu	6	6	6	6
Problém s vlkolakom	2	2	2	2
Vylieč lovca	4	4	4	4

Pri 10 iteráciách ani jedna funkcia nenašla všetky riešenia úlohy Získaj informáciu, ktoré sa našli pri 100 iteráciách. 3. funkcia bola pri 10 iteráciách v prvej úlohe lepšia (našla 6 riešení), ale v druhej našla o jedno riešenie menej ako ostatné funkcie. Keďže používa náhodnosť, môže niekedy skončiť lepšie a niekedy horšie. Pri 20 iteráciách už našli rovnaký počet riešení ako pri 100 iteráciách všetky tri funkcie skoro vo všetkých prípadoch, pričom 2. Funkcia ich našla vždy. A aj keď sú rozdiely výsledkov pri použití rôznych funkcií mále, zdá sa mi, že 2. je predsa len o kúsok lepšia ako ostatné.